Lecture 42 – 50 Questions and Answers

Topic: Advanced Ensemble Learning, Boosting & Model Optimization

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I. Ensemble Learning Concepts (1–10)

1. **Q:** What is the main goal of ensemble learning?
   A: To combine multiple models (weak learners) to improve overall accuracy, robustness, and generalization.

2. **Q:** What are the three major ensemble techniques?
   A: Bagging, Boosting, and Stacking.

3. **Q:** Why do ensembles often outperform individual models?
   A: They reduce variance, bias, and model instability by aggregating predictions from diverse learners.

4. **Q:** What is a "weak learner"?
   A: A model that performs slightly better than random guessing.

5. **Q:** How does combining weak learners create a strong learner?
   A: Through iterative or parallel aggregation, reducing individual errors and enhancing accuracy.

6. **Q:** Which ensemble technique trains models **independently** on bootstrapped data?
   A: Bagging.

7. **Q:** Which ensemble technique trains models **sequentially**, each correcting previous errors?
   A: Boosting.

8. **Q:** What is the key idea behind stacking?
   A: Using a meta-model to combine predictions from multiple base models.

9. **Q:** What type of models are commonly used as base learners in ensemble methods?
   A: Decision Trees.

10. **Q:** What is the primary disadvantage of using multiple models in an ensemble?
    A: Increased computational complexity and reduced interpretability.

II. Boosting Algorithms (11–25)

11. **Q:** What does the term **Boosting** refer to?
 A: Sequentially improving weak learners by focusing more on misclassified samples.

12. **Q:** What does **AdaBoost** stand for?
 A: Adaptive Boosting.

13. **Q:** How does AdaBoost assign sample weights?
 A: It increases the weights of misclassified samples and decreases the weights of correctly classified ones.

14. **Q:** What type of base learners are commonly used in AdaBoost?
 A: Decision stumps (single-split trees).

15. **Q:** How does AdaBoost combine multiple models?
 A: By weighted voting or averaging based on model accuracy.

16. **Q:** What happens to training error as more learners are added in AdaBoost?
 A: It generally decreases rapidly at first and then stabilizes.

17. **Q:** What is **Gradient Boosting**?
 A: A method where new models are trained to predict the residuals (errors) of previous models.

18. **Q:** What is the optimization objective of Gradient Boosting?
 A: To minimize a differentiable loss function, such as MSE or Log Loss.

19. **Q:** Why is Gradient Boosting more flexible than AdaBoost?
 A: Because it can optimize arbitrary differentiable loss functions.

20. **Q:** What is the meaning of “learning rate” in boosting?
 A: A parameter controlling how much each new model contributes to the overall ensemble.

21. **Q:** What happens if the learning rate is too small?
 A: The model learns slowly and requires more iterations.

22. **Q:** What happens if the learning rate is too large?
 A: The model may overfit or fail to converge smoothly.

23. **Q:** What is the function of **early stopping** in boosting?
    A: It halts training when the validation error stops improving to prevent overfitting.

24. **Q:** What does **regularization** do in boosting algorithms?
    A: It penalizes overly complex models to improve generalization.

25. **Q:** Why is boosting sensitive to noisy data?
    A: Because it overemphasizes misclassified samples, including noise.

III. XGBoost, LightGBM, and CatBoost (26–40)

26. **Q:** What does **XGBoost** stand for?
    A: Extreme Gradient Boosting.

27. **Q:** What major advantage does XGBoost offer?
    A: Regularization, parallel computation, and superior efficiency.

28. **Q:** What kind of regularization terms are used in XGBoost?
    A: L1 (Lasso) and L2 (Ridge) regularization.

29. **Q:** How does XGBoost handle missing values?
    A: It automatically learns the optimal direction for missing data during training.

30. **Q:** What is **LightGBM** primarily optimized for?
    A: High performance and low memory usage on large datasets.

31. **Q:** What is the key innovation in LightGBM's tree growth?
    A: Leaf-wise growth instead of level-wise growth.

32. **Q:** What advantage does leaf-wise growth provide?
    A: It reduces loss faster and improves accuracy.

33. **Q:** What is a potential drawback of leaf-wise tree growth?
    A: It can overfit small datasets.

34. **Q:** What is **CatBoost** designed to handle efficiently?
    A: Categorical features without explicit encoding.

35. **Q:** What statistical technique does CatBoost use to avoid overfitting during encoding?
    A: Ordered Target Statistics.

36. **Q:** How does CatBoost reduce prediction bias?
    A: By using symmetric trees and random permutations of data.

37. **Q:** Which boosting algorithm is best suited for high-cardinality categorical data?
    A: CatBoost.

38. **Q:** Which boosting method is known for GPU acceleration support?
    A: XGBoost and LightGBM.

39. **Q:** What is the advantage of using GPU acceleration in boosting?
    A: Significantly faster training for large datasets.

40. **Q:** What is the main trade-off between model complexity and interpretability in boosting algorithms?
    A: Higher complexity yields better accuracy but makes models harder to interpret.

IV. Evaluation and Application (41–50)

41. **Q:** What metric is typically used to evaluate boosting models for regression?
    A: Root Mean Squared Error (RMSE).

42. **Q:** What metric is typically used for classification boosting models?
    A: Log Loss or AUC-ROC.

43. **Q:** Why is cross-validation important in boosting?
    A: To ensure the model generalizes well to unseen data.

44. **Q:** What is the main reason for tuning the number of estimators in boosting?
    A: To balance bias and variance.

45. **Q:** Why are ensemble models preferred in Kaggle competitions?
    A: They achieve higher accuracy by combining diverse learners.

46. **Q:** What visualization helps in interpreting feature importance in boosting models?
    A: Feature importance bar plots or SHAP value plots.

47. **Q:** What does SHAP stand for?
    A: SHapley Additive exPlanations.

48. **Q:** Why is SHAP analysis important?
    A: It explains how each feature contributes to a specific prediction.

49. **Q:** What challenge does boosting face on streaming or real-time data?
    A: Difficulty in incremental updates since boosting models are batch-trained.

50. **Q:** What is one potential improvement to ensemble systems in future AI research?
    A: Integration with neural network ensembles for hybrid deep learning models.